Tutorial 4: FINA2205
Financial Regression Analysis 2

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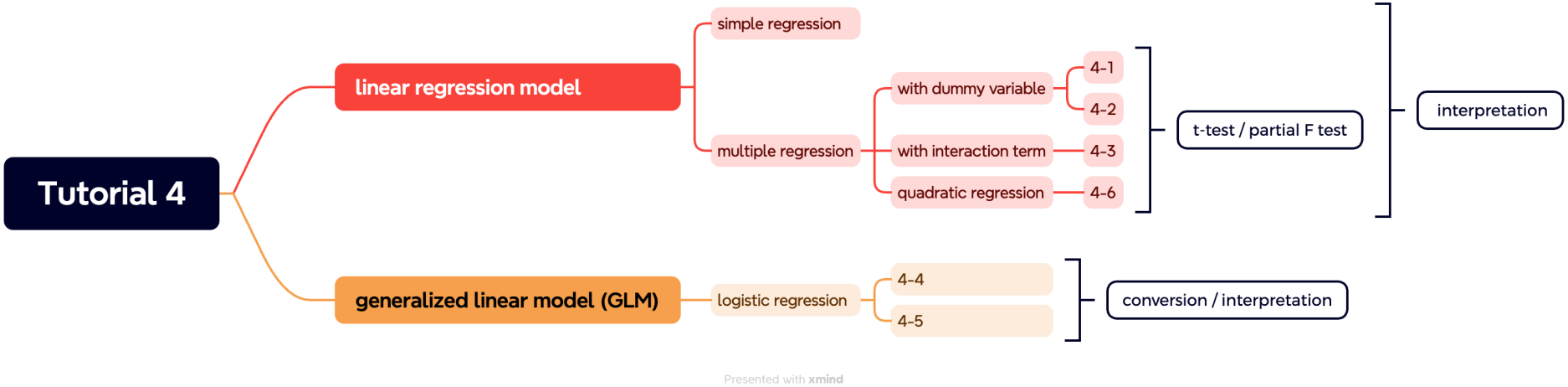
Agenda

- 6 problems



Recap: coefficient interpretation

Q1: Interpret the meaning of slope b_1 (suppose b_1 is a positive number).

- Model: $\hat{Y} = b_0 + b_1 X_1 + b_2 X_2$

1. **Holding constant the effect of $\boldsymbol{X_2}$**, for each <u>increase</u> of one-unit of X_1, Y increase by b_1.
2. For each one-unit increase in X_1, Y will <u>increase</u> an estimated b_1 units, **holding $\boldsymbol{X_2}$ constant**.
3. **For a given $\boldsymbol{X_2}$**, each increase of one-unit X_1 is estimated to result in an <u>increase</u> in Y.

Note: if b_1 is negative, replace "increase" with "decrease".

Recap: t test

Q2: <u>At the 0.05 level of significance</u>, is there any evidence to that variable X_1 makes a significant contribution to the model?

- $\alpha = 0.05$ is our tolerance for error

Step 1: State the hypotheses.

- H_0: $\beta_1 = 0$ (X_1 does not contribute to the model)
- H_1: $\beta_1 \neq 0$ (X_1 contributes to the model)

Note: By convention, the null hypothesis (H_0) always states **"no effect".** The alternative hypothesis (H_1) is what we are trying to find evidence for.

Recap: t test

Step 2: Calculate the t statistic.

$$t_{STAT} = \frac{b_1 - 0}{S_{b_1}}$$

where b_1 is the estimated slope for X_1. And S_{b_1} is the **standard error** of the slope. The 0 is the hypothesized value.

Step 3: Find the critical values.

$$\pm T_{CRIT} = \pm t_{(\alpha/2, df)} = \pm t_{(\alpha/2, n-k-1)}$$

Note1: Because H_1 uses "≠" (not equal), that is a **two-tailed test**. We **split** α into two tails, so each tail has $\frac{\alpha}{2} = \frac{0.05}{2} = 0.025$. That's why we write "±" – there are two critical values, **symmetric** around zero.

Degrees of Freedom	Cumulative Probabilities					
	0.75	0.90	0.95	0.975	0.99	0.995
	Upper-Tail Areas					
	0.25	0.10	0.05	0.025	0.01	0.005
49	0.6795	1.2991	1.6766	2.0096	2.4049	2.6800
50	0.6794	1.2987	1.6759	2.0086	2.4033	2.6778
51	0.6793	1.2984	1.6753	2.0076	2.4017	2.6757
52	0.6792	1.2980	1.6747	2.0066	2.4002	2.6737
53	0.6791	1.2977	1.6741	2.0057	2.3988	2.6718
54	0.6791	1.2974	1.6736	2.0049	2.3974	2.6700
55	0.6790	1.2971	1.6730	2.0040	2.3961	2.6682
56	0.6789	1.2969	1.6725	2.0032	2.3948	2.6665
57	0.6788	1.2966	1.6720	2.0025	2.3936	2.6649
58	0.6787	1.2963	1.6716	2.0017	2.3924	2.6633
59	0.6787	1.2961	1.6711	2.0010	2.3912	2.6618
60	0.6786	1.2958	1.6706	2.0003	2.3901	2.6603
61	0.6785	1.2956	1.6702	1.9996	2.3890	2.6589
62	0.6785	1.2954	1.6698	1.9990	2.3880	2.6575
63	0.6784	1.2951	1.6694	1.9983	2.3870	2.6561
64	0.6783	1.2949	1.6690	1.9977	2.3860	2.6549
65	0.6783	1.2947	1.6686	1.9971	2.3851	2.6536
66	0.6782	1.2945	1.6683	1.9966	2.3842	2.6524
67	0.6782	1.2943	1.6679	1.9960	2.3833	2.6512
68	0.6781	1.2941	1.6676	1.9955	2.3824	2.6501
69	0.6781	1.2939	1.6672	1.9949	2.3816	2.6490
70	0.6780	1.2938	1.6669	1.9944	2.3808	2.6479
71	0.6780	1.2936	1.6666	1.9939	2.3800	2.6469
72	0.6779	1.2934	1.6663	1.9935	2.3793	2.6459
73	0.6779	1.2933	1.6660	1.9930	2.3785	2.6449
74	0.6778	1.2931	1.6657	1.9925	2.3778	2.6439
75	0.6778	1.2929	1.6654	1.9921	2.3771	2.6430
76	0.6777	1.2928	1.6652	1.9917	2.3764	2.6421
77	0.6777	1.2926	1.6649	1.9913	2.3758	2.6412
78	0.6776	1.2925	1.6646	1.9908	2.3751	2.6403
79	0.6776	1.2924	1.6644	1.9905	2.3745	2.6395
80	0.6776	1.2922	1.6641	1.9901	2.3739	2.6387
81	0.6775	1.2921	1.6639	1.9897	2.3733	2.6379
82	0.6775	1.2920	1.6636	1.9893	2.3727	2.6371
83	0.6775	1.2918	1.6634	1.9890	2.3721	2.6364
84	0.6774	1.2917	1.6632	1.9886	2.3716	2.6356
85	0.6774	1.2916	1.6630	1.9883	2.3710	2.6349
86	0.6774	1.2915	1.6628	1.9879	2.3705	2.6342
87	0.6773	1.2914	1.6626	1.9876	2.3700	2.6335
88	0.6773	1.2912	1.6624	1.9873	2.3695	2.6329
89	0.6773	1.2911	1.6622	1.9870	2.3690	2.6322
90	0.6772	1.2910	1.6620	1.9867	2.3685	2.6316
91	0.6772	1.2909	1.6618	1.9864	2.3680	2.6309
92	0.6772	1.2908	1.6616	1.9861	2.3676	2.6303
93	0.6771	1.2907	1.6614	1.9858	2.3671	2.6297
94	0.6771	1.2906	1.6612	1.9855	2.3667	2.6291
95	0.6771	1.2905	1.6611	1.9853	2.3662	2.6286
96	0.6771	1.2904	1.6609	1.9850	2.3658	2.6280
97	0.6770	1.2903	1.6607	1.9847	2.3654	2.6275
98	0.6770	1.2902	1.6606	1.9845	2.3650	2.6269
99	0.6770	1.2902	1.6604	1.9842	2.3646	2.6264
100	0.6770	1.2901	1.6602	1.9840	2.3642	2.6259
110	0.6767	1.2893	1.6588	1.9818	2.3607	2.6213
120	0.6765	1.2886	1.6577	1.9799	2.3578	2.6174
∞	0.6745	1.2816	1.6449	1.9600	2.3263	2.5758
Degrees of Freedom	Upper-Tail Areas					
	0.25	0.10	0.05	0.025	0.01	0.005
	Cumulative Probabilities					
	0.75	0.90	0.95	0.975	0.99	0.995

Recap: t test

Step 4: Compare and conclude.

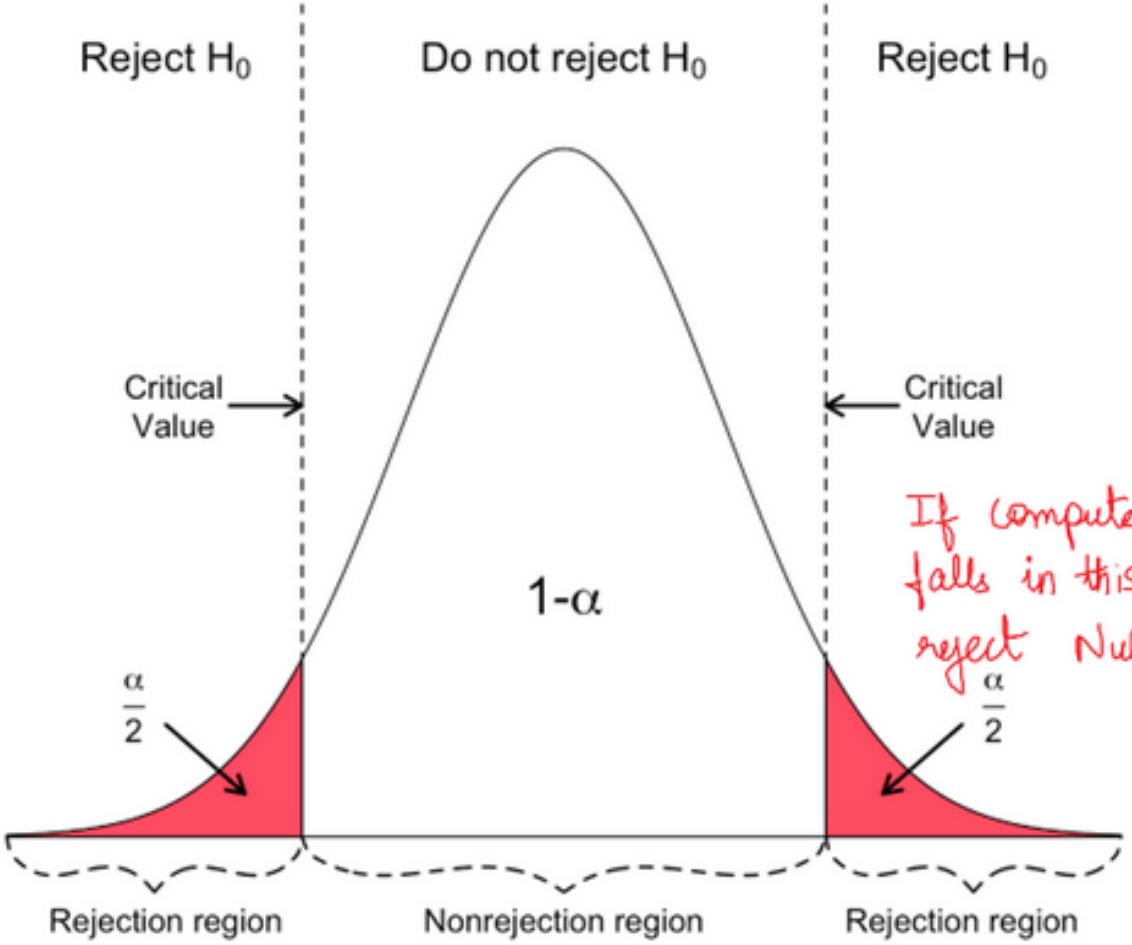
- Scenario 1: reject H_0

"Since $|T_{STAT}| > T_{CRIT}$, the t statistic falls in the rejection region. At the 0.05 level of significance, there is sufficient evidence to conclude that variable X_1 makes a significant contribution to the model."

- Scenario 2: <u>do not</u> reject H_0 ~~Accept H_0~~

"Since $|T_{STAT}| \leq T_{CRIT}$, the t statistic does not fall in the rejection region. At the 0.05 level of significance, there is insufficient evidence to conclude that variable X_1 makes a significant contribution to the model."

t-distribution



If computed t-value falls in this region we reject Null hypothesis

4-1

1. Suppose X_1 is a numerical variable and X_2 is a dummy variable and the regression equation for a sample of $n = 28$ is $\hat{Y}_i = 9 - 5X_{1i} + 3X_{2i}$

a. Interpret the regression coefficient associated with variable X_1.

b. Interpret the regression coefficient associated with variable X_2.

c. Suppose that the t_{STAT} test statistic for testing the contribution of variable X_2 is 1.83. At the 0.05 level of significance, is there evidence that variable X_2 makes a significant contribution to the model?

[1] Click the icon to view the table of critical values of t.

a. Interpret the regression coefficient associated with variable X_1.

Holding constant the effect of (1) _____ for each increase of one unit of (2) _____ Y (3) _____ by ___

b. Interpret the regression coefficient associated with variable X_2.

Holding constant the effect of (4) _____ for each increase of one unit of (5) _____ Y (6) _____ by ___

(1) ○ X_2,	(2) ○ X_2,	(3) ○ increase	(4) ○ X_1,	(5) ○ X_1,	(6) ○ increases
○ X_1,	○ X_1,	○ decreases	○ X_2,	○ X_2,	○ decrease

4-1 solution

a.

- **Holding constant** the effect of (1) X_2, for each increase of one unit of (2) X_1, Y (3) decreases by 5.

b.

- **Holding constant** the effect of (1) X_1, for each increase of one unit of (2) X_2, Y (3) increases by 3.

4-1 (cont.)

c. Suppose that the t_{STAT} test statistic for testing the contribution of variable X_2 is 1.83. At the 0.05 level of significance, is there evidence that variable X_2 makes a significant contribution to the model?

c. Determine the null and alternative hypotheses.

H_0: (7) _____ (8) _____ _____

H_1: (9) _____ (10) _____ _____

(7)
- ○ β_1
- ○ β_0
- ○ β_2

(8)
- ○ $\leq$
- ○ $\geq$
- ○ $=$
- ○ $\neq$

(9)
- ○ β_0
- ○ β_2
- ○ β_1

(10)
- ○ $\neq$
- ○ $<$
- ○ $=$
- ○ $>$

4-1 (cont.)

State the test statistic.

t_{STAT} = _____ (Round to two decimal places as needed.)

Find the critical values.

$\pm t_{\alpha/2} = \pm$ _____ (Round to four decimal places as needed.)

State the conclusion. Choose the correct answer below.

- ○ **A.** Reject H_0. There is sufficient evidence that the variable X_2 makes a significant contribution to the model.
- ○ **B.** Do not reject H_0. There is insufficient evidence that the variable X_2 makes a significant contribution to the model.
- ○ **C.** Do not reject H_0. There is sufficient evidence that the variable X_2 makes a significant contribution to the model.
- ○ **D.** Reject H_0. There is insufficient evidence that the variable X_2 makes a significant contribution to the model.

4-1 solution (cont.)

c.

- H_0: $\beta_2 = 0$
- H_1: $\beta_2 \neq 0$

d.

- $T_{STAT} = 1.83$ (given in question)
- $\pm T_{CRIT} = \pm t_{\frac{\alpha}{2},n-k-1} = \pm t_{\frac{0.05}{2},28-2-1} = \pm t_{0.025,25} = \pm 2.0595$
- $|T_{STAT}| = 1.83 < T_{CRIT}$
- **B.** Do not reject the H_0.

Degrees of Freedom	Cumulative Probabilities					
	0.75	0.90	0.95	0.975	0.99	0.995
	Upper-Tail Areas					
	0.25	0.10	0.05	0.025	0.01	0.005
1	1.0000	3.0777	6.3138	12.7062	31.8207	63.6574
2	0.8165	1.8856	2.9200	4.3027	6.9646	9.9248
3	0.7649	1.6377	2.3534	3.1824	4.5407	5.8409
4	0.7407	1.5332	2.1318	2.7764	3.7469	4.6041
5	0.7267	1.4759	2.0150	2.5706	3.3649	4.0322
6	0.7176	1.4398	1.9432	2.4469	3.1427	3.7074
7	0.7111	1.4149	1.8946	2.3646	2.9980	3.4995
8	0.7064	1.3968	1.8595	2.3060	2.8965	3.3554
9	0.7027	1.3830	1.8331	2.2622	2.8214	3.2498
10	0.6998	1.3722	1.8125	2.2281	2.7638	3.1693
11	0.6974	1.3634	1.7959	2.2010	2.7181	3.1058
12	0.6955	1.3562	1.7823	2.1788	2.6810	3.0545
13	0.6938	1.3502	1.7709	2.1604	2.6503	3.0123
14	0.6924	1.3450	1.7613	2.1448	2.6245	2.9768
15	0.6912	1.3406	1.7531	2.1315	2.6025	2.9467
16	0.6901	1.3368	1.7459	2.1199	2.5835	2.9208
17	0.6892	1.3334	1.7396	2.1098	2.5669	2.8982
18	0.6884	1.3304	1.7341	2.1009	2.5524	2.8784
19	0.6876	1.3277	1.7291	2.0930	2.5395	2.8609
20	0.6870	1.3253	1.7247	2.0860	2.5280	2.8453
21	0.6864	1.3232	1.7207	2.0796	2.5177	2.8314
22	0.6858	1.3212	1.7171	2.0739	2.5083	2.8188
23	0.6853	1.3195	1.7139	2.0687	2.4999	2.8073
24	0.6848	1.3178	1.7109	2.0639	2.4922	2.7969
25	0.6844	1.3163	1.7081	2.0595	2.4851	2.7874
26	0.6840	1.3150	1.7056	2.0555	2.4786	2.7787
27	0.6837	1.3137	1.7033	2.0518	2.4727	2.7707
28	0.6834	1.3125	1.7011	2.0484	2.4671	2.7633
29	0.6830	1.3114	1.6991	2.0452	2.4620	2.7564
30	0.6828	1.3104	1.6973	2.0423	2.4573	2.7500
31	0.6825	1.3095	1.6955	2.0395	2.4528	2.7440
32	0.6822	1.3086	1.6939	2.0369	2.4487	2.7385
33	0.6820	1.3077	1.6924	2.0345	2.4448	2.7333
34	0.6818	1.3070	1.6909	2.0322	2.4411	2.7284
35	0.6816	1.3062	1.6896	2.0301	2.4377	2.7238
36	0.6814	1.3055	1.6883	2.0281	2.4345	2.7195
37	0.6812	1.3049	1.6871	2.0262	2.4314	2.7154
38	0.6810	1.3042	1.6860	2.0244	2.4286	2.7116
39	0.6808	1.3036	1.6849	2.0227	2.4258	2.7079
40	0.6807	1.3031	1.6839	2.0211	2.4233	2.7045

Recap: Dummy variables

- A dummy variable (D) is a categorical independent variable.

$$\hat{Y} = b_0 + b_1 X_1 + b_2 D_2$$

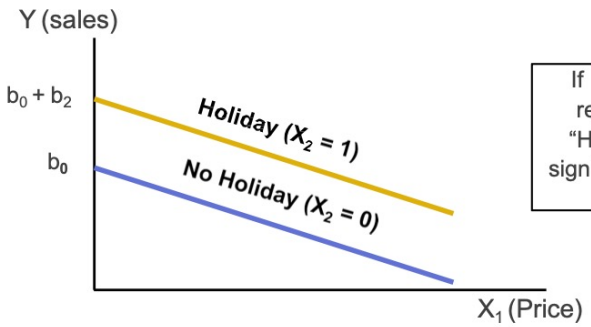
- **How to interpret $\boldsymbol{b_2}$**: it represents the difference in predicted Y between the two groups ($D = 1\ vs.\ D = 0$), holding X_1 constant.
- When $D = 0, \hat{Y} = b_0 + b_1 X_1$
- When $D = 1, \hat{Y} = (b_0 + b_2) + b_1 X_1$
- So b_2 is the **shift** in the intercept – same slope, different starting point.
- If more than two levels, the number of dummy variables needed is ($number\ of\ levels - 1$).

$$\hat{Y} = b_0 + b_1X_1 + b_2X_2$$

Let: Y = pie sales
X_1 = price
X_2 = holiday ($X_2 = 1$ if a holiday occurred during the week)
($X_2 = 0$ if there was no holiday that week)

$$\hat{Y} = b_0 + b_1X_1 + b_2(1) = (b_0 + b_2) + b_1X_1$$ **Holiday**

$$\hat{Y} = b_0 + b_1X_1 + b_2(0) = (b_0) + b_1X_1$$ **No Holiday**



If H_0: $\beta_2 = 0$ is rejected, then "Holiday" has a significant effect on pie sales

Example:

$$\widehat{\text{Sales}} = 300 - 30(\text{Price}) + 15(\text{Holiday})$$

Sales: number of pies sold per week
Price: pie price in $

Holiday: $\begin{cases} 1 & \text{If a holiday occurred during the week} \\ 0 & \text{If no holiday occurred} \end{cases}$

$b_2 = 15$: on average, sales were 15 pies greater in weeks with a holiday than in weeks without a holiday, given the same price

Recap: Dummy variables

When to use what method?

- ANOVA (Analysis of Variance) tests whether the means of **different groups** are significantly different from each other.
- ANOVA → **all** independent variables are **categorical**.
- Multiple regression → independent variables include **numerical** variables (you need to model a continuous relationship)
- In the pie sales example, we use multiple regression rather than ANOVA because the independent variables include both a numerical variable (*Price*) and a categorical variable (*Holiday*).

Recap: Dummy variables

How to test whether the dummy term is significant?

- Two options: t test or partial F test

t test – Does one variable make a significant contribution?

- → Test each **variable** already in the model, one at a time.

Partial F test – Does adding a new variable significantly improve the model? (not examinable)

- → Compare two **models**: one without the new variable vs. one with it.

4-2

2. The chair of the accounting department plans to develop a regression model to predict the grade point average in accounting for those students who are graduating and have completed the accounting major, based on a student's SAT score and whether the student received a grade of B or higher in the introductory statistics course (0 = no and 1 = yes). Complete parts (a) and (b) below.

a. Explain the steps involved in developing a regression model for these data. Be sure to indicate the particular models you need to evaluate and compare. Choose the correct answers below.

Let Y be the student's grade point average in accounting, X_1 be the student's SAT score, and X_2 be whether the student received a grade of B or higher in introductory statistics. Use technology to run a (1) _____ on the data, and determine whether each variable makes a significant contribution using the (2) _____ Lastly, add an interaction variable to the regression, and use the (3) _____ to determine whether this new variable significantly improves the model. Variables that do not make a significant contribution should be removed from the model.

(1)
- ○ multiple regression
- ○ two-way ANOVA

(2)
- ○ chi-square test statistic.
- ○ t_{STAT} test statistic.

(3)
- ○ partial F test
- ○ t_{STAT} test statistic

4-2

b. Suppose the regression coefficient for the variable whether the student received a grade of B or higher in the introductory statistics course is + 0.30. How do you interpret this result?

- **A.** Of students who are graduating and have completed the accounting major, those who got a B or higher in the introductory statistics course generally have a lower GPA in accounting than those who did not get a B or better in the introductory statistics course.
- **B.** Students who got a B or higher in the introductory statistics course generally have a lower GPA in accounting than those who did not.
- **C.** With SAT score held constant, it is estimated that the predicted grade point average in accounting is 0.30 greater for a student who received a grade of B or higher in the introductory statistics course than those who did not.
- **D.** Students who got a B or higher in the introductory statistics course generally have a higher GPA in accounting than those who did not.

? SAT

4-2

a.

- Use technology to run a (1) multiple regression on the data, and determine whether each variable makes a significant contribution using the (2) t-test statistic. Use the (3) partial F test (not examinable) to determine whether this new variable significantly improves the model.

b.

- C. **With SAT score held constant,** it is estimated that the predicted GPA is 0.30 greater for a student who received a grade…than those who did not.

Recap: Interaction terms

An interaction term captures whether the effect of one variable on *Y* **depends on** the level of another variable.

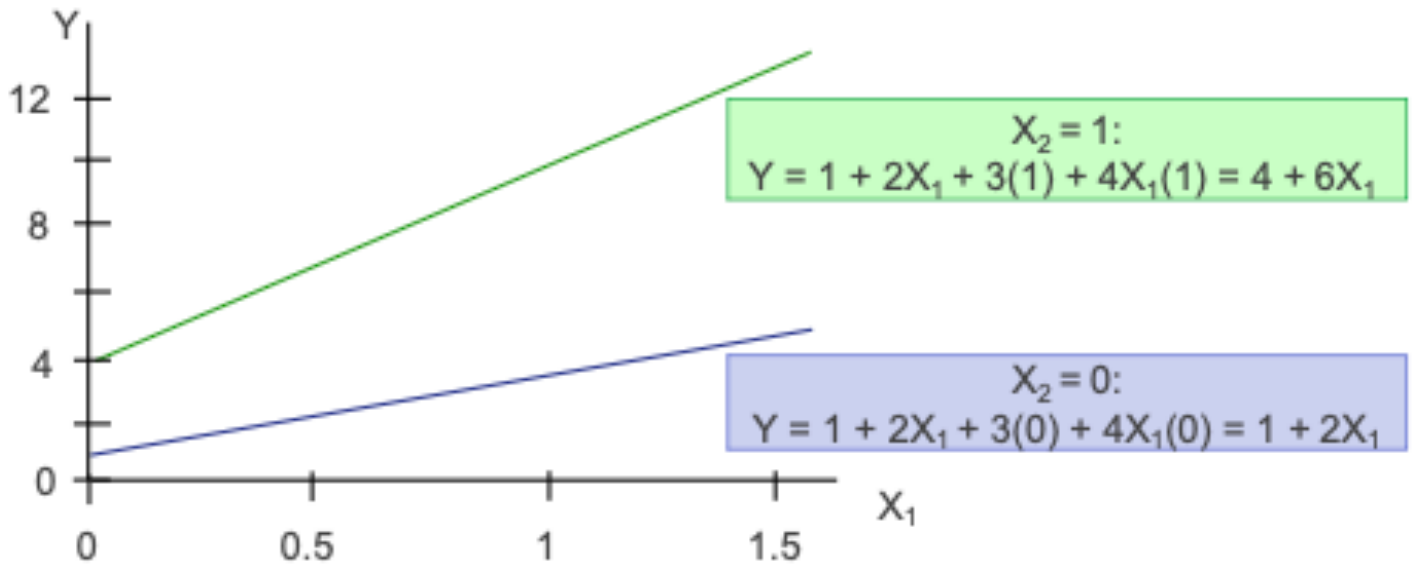
Original model: $\hat{Y} = b_0 + b_1 X_1 + b_2 D_2$

Model with interaction: $\hat{Y} = b_0 + b_1 X_1 + b_2 D_2 + b_3 (X_1 \times D_2)$

- $\rightarrow \hat{Y} = b_0 + (b_1 + b_3 D_2) X_1 + b_2 D_2$
- **How to interpret $\boldsymbol{b_3}$**: If $b_3 = 0 \rightarrow$ the effect of X_1 on *Y* is the same regardless of D_2. If $b_3 \neq 0 \rightarrow$ the effect of X_1 on *Y* **changes** depending on the value of X_2.

- Suppose X_2 is a dummy variable and the estimated regression equation is:

$$\hat{Y} = 1 + 2X_1 + 3X_2 + 4X_1X_2$$



Slopes are different if the effect of X_1 on Y depends on X_2 value

4-3

3. For the data in the following table, radio advertising and newspaper advertising were used to predict sales, resulting in the equation $Y = 164.152 + 12.902X_{1i} + 16.725X_{2i}$, where X_1 represents radio advertising and X_2 represents newspaper advertising. Develop a regression model to predict sales that includes radio advertising, newspaper advertising, and the interaction of radio advertising and newspaper advertising.
[2] Click the icon to view the table.
   a. At the 0.05 level of significance, is there evidence that the interaction term makes a significant contribution to the model?
   b. Which regression model is more appropriate, the one given above or the one developed in this problem?

a. Determine the null and alternative hypotheses. Let X_3 be the interaction of X_1 and X_2, defined as $X_3 = X_1 \times X_2$, and β_3 be the slope of Y with variable X_3.

H_0: β_3 (1) _____ _____

H_1: β_3 (2) _____ _____

4-3 (cont.)

Beyond the scope

Determine the partial F_{STAT} test statistic.

F_{STAT} = _____ (Round to two decimal places as needed.)

Determine the p-value.

p-value = _____ (Round to three decimal places as needed.)

Draw an appropriate conclusion.

(3) _____ the null hypothesis. Conclude that there (4) _____ evidence that the interaction term makes a significant contribution to the model.

b. Which regression model is more appropriate, the one given above or the one developed in this problem? Choose the correct answer below.

- ○ Use the model developed in this problem.
- ○ Use the model given in the problem statement.

4-3 solution

a.

- H_0: $\beta_3 = 0$
- H_1: $\beta_3 \neq 0$

b.

$$F_{STAT} = \frac{(SSR(full) - SSR(reduced)/m}{MSE(full)}$$

$All\ except\ new\ set\ of\ m\ variables$

where *m* is the number of variables added simultaneously to the model.

not examinable

4-3 solution (cont.)

not examinable

- From PHSstat output
- $F_{STAT} = \frac{(2231948.7778-2003645.5811)/1}{14799.7270} = 15.4262$
- Use **F.DIST.RT** function on excel. $P - value = \mathbf{0.001 < 0.05}$

We (3) <u>reject</u> the null hypothesis, as there is (4) <u>sufficient</u> evidence to conclude that the interaction term makes contributions to the model

b. <u>Use the model developed in this problem</u> (the interaction term should be retained).

City	Sales ($Thousands)	Radio Advertising ($Thousands)	Newspaper Advertising ($Thousands)
1	965	0	39
2	1,121	0	40
3	878	26	26
4	628	25	24
5	911	31	29
6	975	31	31
7	938	35	34
8	1,171	37	37
9	883	40	24
10	988	40	26
11	1,629	45	43
12	1,567	44	46
13	1,044	53	0
14	920	48	0
15	1,321	52	25
16	1,325	58	25
17	1,409	62	30
18	1,435	58	29
19	1,526	65	36
20	1,751	63	34
21	1,870	70	41
22	1,712	73	40

Recap: quadratic regression

What is it?

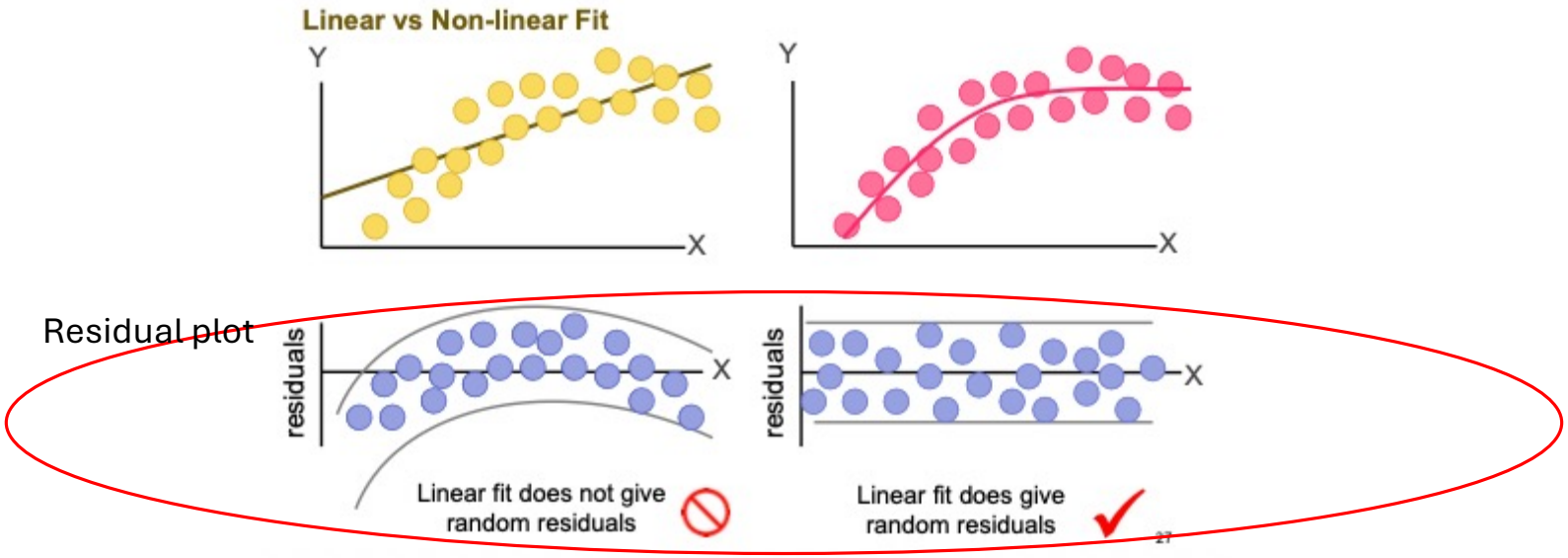
- Adding a squared term to capture a **curved relationship**.

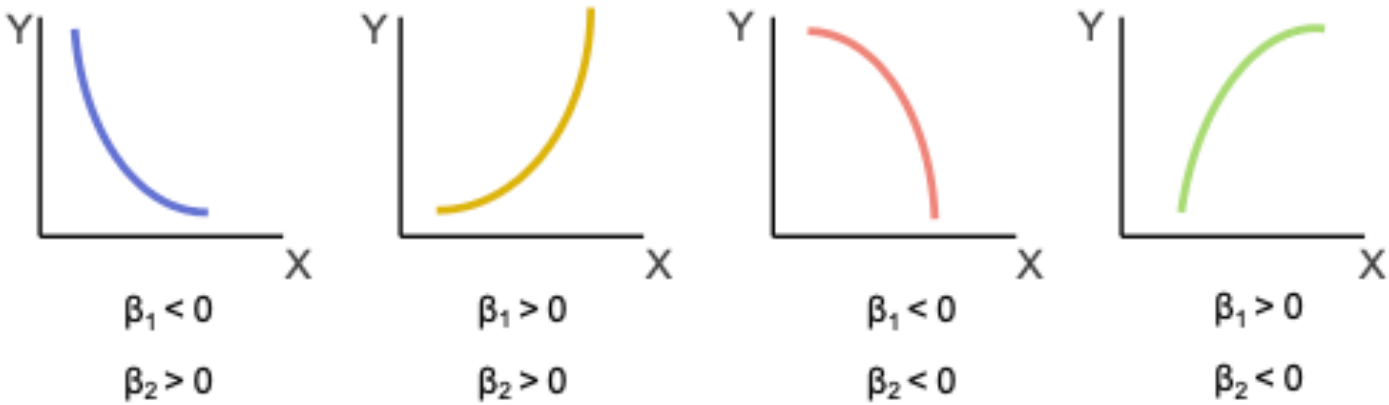
Original model: $\hat{Y} = b_0 + b_1 X_1$

Model with interaction: $\hat{Y} = b_0 + b_1 X_1 + b_2 X_1^2$

When do we need it?

When the <u>residual plot</u> of the linear model shows a special shape or curved pattern → the relationship is not a straight line → add X_1^2 to capture the curve.


Linear vs Non-linear Fit
Y
X
Y
X
Residual plot
residuals
X
residuals
X
Linear fit does not give random residuals
Linear fit does give random residuals


Y
X
$\beta_1 < 0$
$\beta_2 > 0$
Y
X
$\beta_1 > 0$
$\beta_2 > 0$
Y
X
$\beta_1 < 0$
$\beta_2 < 0$
Y
X
$\beta_1 > 0$
$\beta_2 < 0$

Recap: quadratic regression

The key question: Does adding X_1^2 significantly improve the model?

This is **the same logic** as testing an interaction term:

	Interaction term	Quadratic term
What are we adding?	$X_1 \times D_2$	X_1^2
What are we testing?	$\beta_3 = 0?$	$\beta_2 = 0?$
Which test?	t test or partial F test	t test or partial F test

Recap: quadratic regression

- H_0: $\beta_2 = 0$ (quadratic term does not improve the model)
- H_1: $\beta_2 \neq 0$ (quadratic term improves the model)
- If reject H_0 $\rightarrow$ use the quadratic model.
- If do not reject H_0 $\rightarrow$ drop X_1^2, use the linear model only.

4-6

6. The quadratic regression equation shown below is for a sample of $n = 25$. Complete parts a through d.

$$\hat{Y}_i = 6 + 8X_{1i} + 7.5X_{1i}^2$$

a. Predict Y for $X_1 = 4$.

$\hat{Y} =$ _____

(Type an integer or a decimal.)

b. Suppose that the computed t_{STAT} test statistic for the quadratic regression coefficient is 2.64. At the 0.02 level of significance, is there evidence that the quadratic model is better than the linear model?

Choose the correct null and alternative hypotheses below.

- ○ **A.** H_0: Including the quadratic effect does not significantly improve the model ($\beta_2 = 0$).
  H_1: Including the quadratic effect significantly improves the model ($\beta_2 \neq 0$).
- ○ **B.** H_0: There is no overall relationship between X and Y ($\beta_1 = \beta_2 = 0$).
  H_1: There is an overall relationship between X and Y ($\beta_1 \neq 0$ or $\beta_2 \neq 0$).
- ○ **C.** H_0: There is an overall relationship between X and Y ($\beta_1 \neq 0$ or $\beta_2 \neq 0$).
  H_1: There is no overall relationship between X and Y ($\beta_1 = \beta_2 = 0$).
- ○ **D.** H_0: Including the quadratic effect significantly improves the model ($\beta_2 \neq 0$).
  H_1: Including the quadratic effect does not significantly improve the model ($\beta_2 = 0$).

4-6 solution

a.

- $\hat{Y} = 6 + 8\times4 + 7.5\times4^2 = 158$

b.

A.

$$H_0\colon\ \beta_2 = 0$$
$$H_1\colon\ \beta_2 \neq 0$$

Note: *the problem with Option B is that it's a* ***joint*** *hypothesis. If we reject* $H_0\colon \beta_1 = \beta_1 = 0$*, it only tells us that at least one of them is non-zero.* ***It doesn't tell us which one.***

4-6 (cont.)

b. Suppose that the computed t_{STAT} test statistic for the quadratic regression coefficient is 2.64. At the 0.02 level of significance, is there evidence that the quadratic model is better than the linear model?

Determine the critical value(s). Select the correct choice below, and, if necessary, fill in the answer box to complete your choice.

(Round to three decimal places as needed.)

- **A.** The critical values are ± _____.
- ~~**B.** The critical value is _____.~~

Choose the correct conclusion below.

- **A.** Reject H_0. The quadratic term is not significant.
- **B.** Do not reject H_0. The quadratic term is significant.
- **C.** Do not reject H_0. The quadratic term is not significant.
- **D.** Reject H_0. The quadratic term is significant.

4-6 solution (cont.)

- $T_{STAT} = 2.64$ (given in question)
- $T_{CRIT} = \pm t_{\frac{\alpha}{2},n-k-1} = \pm\, t_{\frac{0.02}{2},25-2-1} = \pm t_{0.01,22} = \pm\, 2.5083$
- $|T_{STAT}| > T_{CRIT} = 2.5083$
- D. Reject H_0. The quadratic term is significant.

Degrees of Freedom	Cumulative Probabilities					
	0.75	0.90	0.95	0.975	0.99	0.995
	Upper-Tail Areas					
	0.25	0.10	0.05	0.025	0.01	0.005
1	1.0000	3.0777	6.3138	12.7062	31.8207	63.6574
2	0.8165	1.8856	2.9200	4.3027	6.9646	9.9248
3	0.7649	1.6377	2.3534	3.1824	4.5407	5.8409
4	0.7407	1.5332	2.1318	2.7764	3.7469	4.6041
5	0.7267	1.4759	2.0150	2.5706	3.3649	4.0322
6	0.7176	1.4398	1.9432	2.4469	3.1427	3.7074
7	0.7111	1.4149	1.8946	2.3646	2.9980	3.4995
8	0.7064	1.3968	1.8595	2.3060	2.8965	3.3554
9	0.7027	1.3830	1.8331	2.2622	2.8214	3.2498
10	0.6998	1.3722	1.8125	2.2281	2.7638	3.1693
11	0.6974	1.3634	1.7959	2.2010	2.7181	3.1058
12	0.6955	1.3562	1.7823	2.1788	2.6810	3.0545
13	0.6938	1.3502	1.7709	2.1604	2.6503	3.0123
14	0.6924	1.3450	1.7613	2.1448	2.6245	2.9768
15	0.6912	1.3406	1.7531	2.1315	2.6025	2.9467
16	0.6901	1.3368	1.7459	2.1199	2.5835	2.9208
17	0.6892	1.3334	1.7396	2.1098	2.5669	2.8982
18	0.6884	1.3304	1.7341	2.1009	2.5524	2.8784
19	0.6876	1.3277	1.7291	2.0930	2.5395	2.8609
20	0.6870	1.3253	1.7247	2.0860	2.5280	2.8453
21	0.6864	1.3232	1.7207	2.0796	2.5177	2.8314
22	0.6858	1.3212	1.7171	2.0739	2.5083	2.8188
23	0.6853	1.3195	1.7139	2.0687	2.4999	2.8073
24	0.6848	1.3178	1.7109	2.0639	2.4922	2.7969
25	0.6844	1.3163	1.7081	2.0595	2.4851	2.7874
26	0.6840	1.3150	1.7056	2.0555	2.4786	2.7787
27	0.6837	1.3137	1.7033	2.0518	2.4727	2.7707
28	0.6834	1.3125	1.7011	2.0484	2.4671	2.7633
29	0.6830	1.3114	1.6991	2.0452	2.4620	2.7564
30	0.6828	1.3104	1.6973	2.0423	2.4573	2.7500
31	0.6825	1.3095	1.6955	2.0395	2.4528	2.7440
32	0.6822	1.3086	1.6939	2.0369	2.4487	2.7385
33	0.6820	1.3077	1.6924	2.0345	2.4448	2.7333
34	0.6818	1.3070	1.6909	2.0322	2.4411	2.7284
35	0.6816	1.3062	1.6896	2.0301	2.4377	2.7238
36	0.6814	1.3055	1.6883	2.0281	2.4345	2.7195
37	0.6812	1.3049	1.6871	2.0262	2.4314	2.7154
38	0.6810	1.3042	1.6860	2.0244	2.4286	2.7116
39	0.6808	1.3036	1.6849	2.0227	2.4258	2.7079
40	0.6807	1.3031	1.6839	2.0211	2.4233	2.7045

4-6 (cont.)

c. Suppose that the computed t_{STAT} test statistic for the quadratic regression coefficient is 1.25. At the 0.02 level of significance, is there evidence that the quadratic model is better than the linear model?

- **A.** Do not reject H_0. The quadratic term is significant.
- **B.** Do not reject H_0. The quadratic term is not significant.
- **C.** Reject H_0. The quadratic term is not significant.
- **D.** Reject H_0. The quadratic term is significant.

d. Suppose the regression coefficient for the linear effect is -8.0. Predict Y for $X_1 = 4$.

$\hat{Y} =$ _____

(Type an integer or a decimal.)

4-6 solution (cont.)

c.

- $T_{STAT} = 1.25$ (supposed)
- $T_{CRIT} = \pm 2.5083$
- $|T_{STAT}| < T_{CRIT} = 2.5083$
- B. Do not Reject H_0. The quadratic term is not significant.

d.

The quadratic regression equation shown below is for a sample of n = 25. Complete parts a through d.

$$\hat{Y}_i = 6 + 8X_{1i} + 7.5X_{1i}^2$$

- $\hat{Y} = 6 - 8X_1 + 7.5X_1^2 = 6 - 8\times 4 + 7.5\times(4)^2 = 94$

Recap: Logistic regression

When to use?

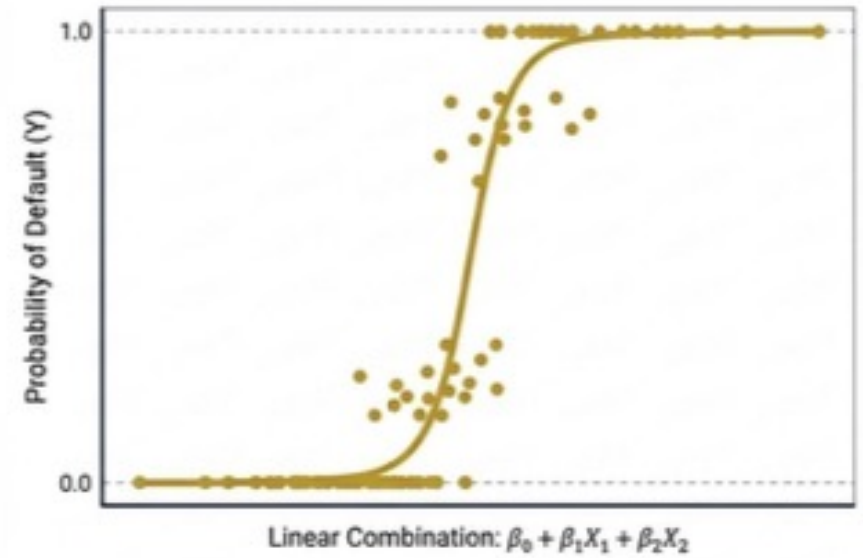
- When Y is binary (only two outcomes: 0 or 1, e.g., pass/fail, yes/no).
- The model: $\ln(\textit{estimated odds ratio}) = b_0 + b_1 X_1 + b_2 X_2$

Why ln(odds)?

- We don't just want 0 or 1 – we need probability to know how confident we are – 0.99 and 0.51 are both "1", but they are very different decisions.

Recap: Logistic regression

- But probability is bounded between 0 and 1 – a linear equation can produce values outside this range.
- So we convert to $odds\ ratio = \frac{P}{1-P}$ – now the range is 0 to +∞. But a linear equation can be negative.
- So we take $\ln(odds\ ratio)$ – now the range is −∞ to +∞.



Recap: Logistic regression

Three key terms: know how to convert between them

- ln(odds) → odds → probability
- Step 1: $\ln(estimated\ odds\ ratio)$ – plug X values into the equation
- Step 2: Estimated $\mathrm{O}dds\ ratio = e^{\ln(estimated\ odds\ ratio)}$
- Step 3:

$$\mathrm{E}stimated\ Probability\ of\ an\ event\ (P) = \frac{estimated\ odds\ ratio}{1 + estimated\ odds\ ratio}$$

- This model predicts the probability of the event of interest ($Y = 1$).

Recap: Logistic regression

How to interpret coefficients?

$$\ln(\textit{estimated odds ratio}) = b_0 + b_1 X_1 + b_2 X_2$$

- Same pattern as multiple regression, but replace "*Y*" with "the natural logarithm of odds ratio,"
- Holding X_2 constant, for each one-unit increase in X_1, <u>the natural logarithm of the odds ratio</u> is estimated to increase by a mean of b_1.

4-4

4. Given an estimated odds ratio of 12.5, find the estimated probability of an event of interest.

   The estimated probability of an event of interest is _____.
   (Round to four decimal places as needed.)

4-4 solution

$$Estimated\ probability\ of\ event = \frac{estimated\ odds\ ratio}{1 + estimated\ odds\ ratio}$$

- $P = \frac{12.5}{1+12.5} = 0.9259$
- This model predicts a 92.59% probability of the event of interest.

4-5

5. Consider the following logistic regression equation. Complete parts (a) through (c).

$$\ln(\text{estimated odds ratio}) = 0.2 + 0.5X_{1i} + 0.4X_{2i}$$

a. Interpret the meaning of the logistic regression coefficients.

Holding constant the effects of X_2, for each additional unit of X_1 the natural logarithm of the odds ratio is estimated to increase by a mean of (1) _____. Holding constant the effects of X_1, for each additional unit of X_2 the natural logarithm of the odds ratio is estimated to increase by a mean of (2) _____.

(1)
- ○ 0.4
- ○ 0.6
- ○ 0.2
- ○ 0.5
- ○ 0.7

(2)
- ○ 0.2
- ○ 0.7
- ○ 0.5
- ○ 0.4
- ○ 0.6

4-5

b. If $X_1 = 2$ and $X_2 = 3.5$, find the estimated odds ratio.

Estimated odds ratio = _____
(Round to four decimal places as needed.)

c. On the basis of the results of (b), compute the estimated probability of an event of interest.

Estimated probability of success = _____
(Round to four decimal places as needed.)

4-5 solution

a.

- **Holding constant** the effects of X_2…the natural logarithm of the odds ratio is estimated to **increase** by a mean of (1) 0.5. **Holding constant** the effects of X_1… to **increase** by a mean of (2) 0.4.

b.

- $\hat{Y} = Ln(estimated\ odds\ ratio) = 0.2 + 0.5X_{1i} + 0.4X_{2i} = 0.2 + 0.5\times\mathbf{2} + 0.4\times\mathbf{3.5} = 2.6$
- $Estimated\ odds\ ratio = e^{2.6} = 13.4637$

4-5 solution

c.

$$Estimated\ probability\ of\ event\ of\ interest = \frac{Estimated\ odds\ ratio}{1 + Estimated\ odds\ ratio}$$

- On the basis of the results of (b),
- $Estimated\ probability\ of\ success = \frac{13.4637}{1+13.4637} = 0.9309$
- This model predicts a 93.09% probability of the event of interest.